

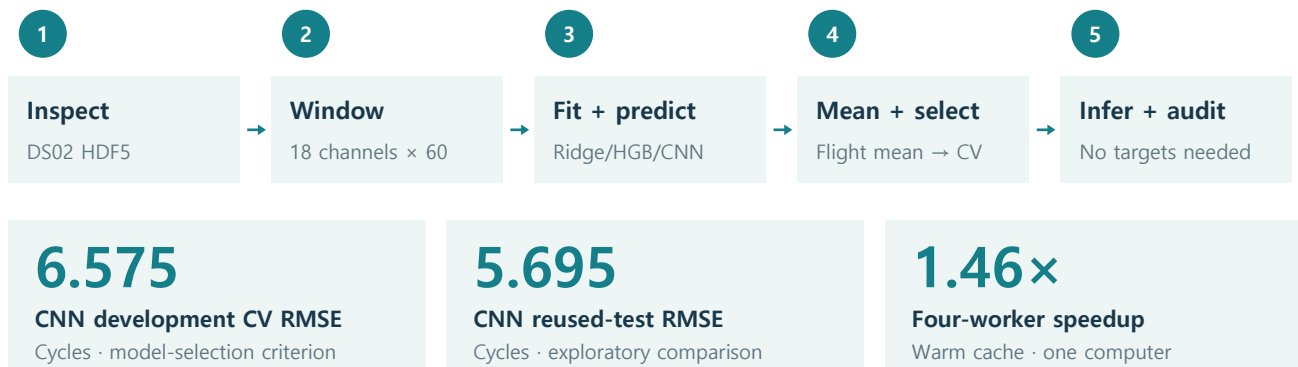
Aircraft Engine Remaining Useful Life Prediction

NASA N-CMAPSS DS02 Project Report

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Objective and completed scope

The project predicts engine remaining useful life (RUL) from large HDF5 histories and measures parallel-processing costs. Three baselines, 24 window/policy candidates, a 1D CNN and saved-model inference were implemented on 2 October. All reported numbers come from actual saved experiments.



Development CV selected Ridge originally and CNN after extension. CNN reused-test MAE is 4.521 cycles. Extension design followed test inspection, so test differences remain exploratory. The fastest preprocessing median was 1.473 s (120,000-row chunks); every run had the same feature fingerprint.

Inspected data

Split	Engine IDs	Raw rows	Cycles	Windows
Development	2, 5, 10, 16, 18, 20	5,263,447	446	87,510
Test	11, 14, 15	1,253,743	202	20,796

Total: 6,517,190 rows from nine simulated engines. All development engines have flight class 3; test engines 11, 14 and 15 have classes 3, 1 and 2 respectively. A large row count does not increase the number of independent engines.

Acquisition provenance

The original sources are NASA PCoE entry 17 [1] and the N-CMAPSS paper [2]. The analyzed DS02 file came from the third-party Figshare mirror v1 [3], containing 2,450,472,504 bytes. Its published MD5 was verified. The NASA full ZIP is 15,760,443,389 bytes, not the DS02 file size; byte equivalence with NASA's archive was not independently established.

Baseline Methods and Validation

Observable inputs and features

Inputs were four operating-condition channels W and fourteen measured-sensor channels X_s . Degradation variables T , health state hs , virtual sensors X_v , flight class, engine ID and cycle ID were excluded as predictors. IDs served only for boundaries, splits and aggregation; Y served only as the target.

One engine, one recorded flight

1-60	61-120	121-180	tail	18 channels × 5 stats = 90 features mean / std / min / max / slope
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Window = 60, stride = 60; drop incomplete tail; reset at next flight

Windows never cross engine/cycle boundaries, while state persists across read chunks. Standard deviation uses `ddof=0`; slope is linear change per sample. Calculations use float64 and stored features use float32.

Incomplete tails excluded 12,847 development and 5,983 test samples. Neither split lost windows to non-finite values. Initial extraction with four workers took 1.466 s for development and 1.112 s for test. These single-run times are distinct from repeated benchmark medians.

Models and fitting

Model	Fixed settings
Mean baseline	Weighted mean of fitting targets
Ridge	StandardScaler; alpha=10.0
HGB	max_iter=100; max_leaf_nodes=15; l2_regularization=1.0; early_stopping=False; seed=410

Within each fitting partition, engines received equal total weight and cycles within each engine received equal total weight. Relative window weights were $1 / (\text{windows in its cycle} \times \text{cycles in its engine})$, normalized to mean 1. Ridge scaling and regression were both weighted and fitted on the current training fold only.

Engine-disjoint cross-validation

Engine split / folds	2	5	10	16	18	20	11	14	15
	C3	C3	C3	C3	C3	C3	C3	C1	C2
Fold 1	Fit	CV	Fit	CV	Fit	Fit	Test	Test	Test
Fold 2	Fit	Fit	CV	Fit	Fit	CV	Test	Test	Test
Fold 3	CV	Fit	Fit	Fit	CV	Fit	Test	Test	Test

C = flight class. Test engines are excluded from fitting and selection.

Each development engine was held out once in three GroupKFold folds. Window predictions were averaged within each engine and flight cycle to estimate RUL at flight end; cycle targets were checked for consistency. Cycle-level MAE and RMSE were computed per engine, then averaged equally across engines. The lowest development macro RMSE selected the model. All candidates were refitted on six development engines and evaluated once on the fixed three-engine test split. Targets and predictions were not clipped or transformed.

Preserved Baseline Results

Errors are in remaining flight cycles. Macro metrics below are unweighted means of engine-level cycle metrics. Development scores use out-of-fold predictions for all six engines; test scores use 202 cycles from three independent engines.

Model	CV MAE	CV RMSE	Test MAE	Test RMSE
Mean baseline	18.851	21.914	17.235	20.104
Ridge *	9.594	11.383	9.878	11.276
HGB	9.749	11.775	9.755	11.227

* Model selected by development cross-validation. HGB denotes HistGradientBoostingRegressor. Improvement over the baseline is specific to this split; it is not a significance claim or a guarantee for other engine populations.

Per-engine test results

Model	Engine	Class	Cycles	MAE	RMSE
Mean baseline	11	3	59	15.746	18.676
Mean baseline	14	1	76	19.009	21.953
Mean baseline	15	2	67	16.950	19.684
Ridge	11	3	59	10.003	11.417
Ridge	14	1	76	10.739	12.296
Ridge	15	2	67	8.894	10.116
HGB	11	3	59	8.749	10.081
HGB	14	1	76	12.088	13.931
HGB	15	2	67	8.429	9.670

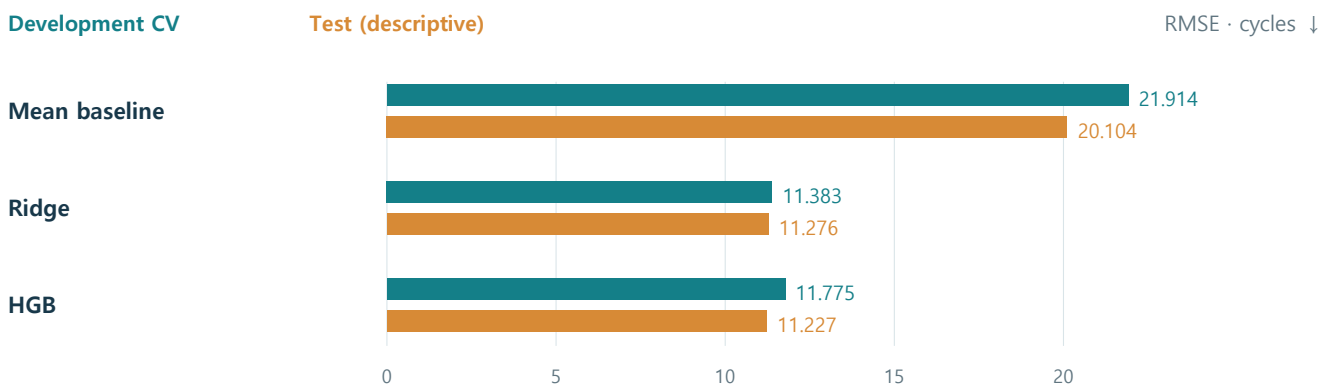


Figure 1. Development/test RMSE after flight-end aggregation. All bars start at zero. Selection used development cross-validation, independently of test ranking.

Ridge had lower development RMSE, while HGB had 0.049 cycles lower test RMSE. Test outcomes did not revise the original baseline's selection rules or settings; its selected model remains Ridge.

Baseline Trajectories and Limits

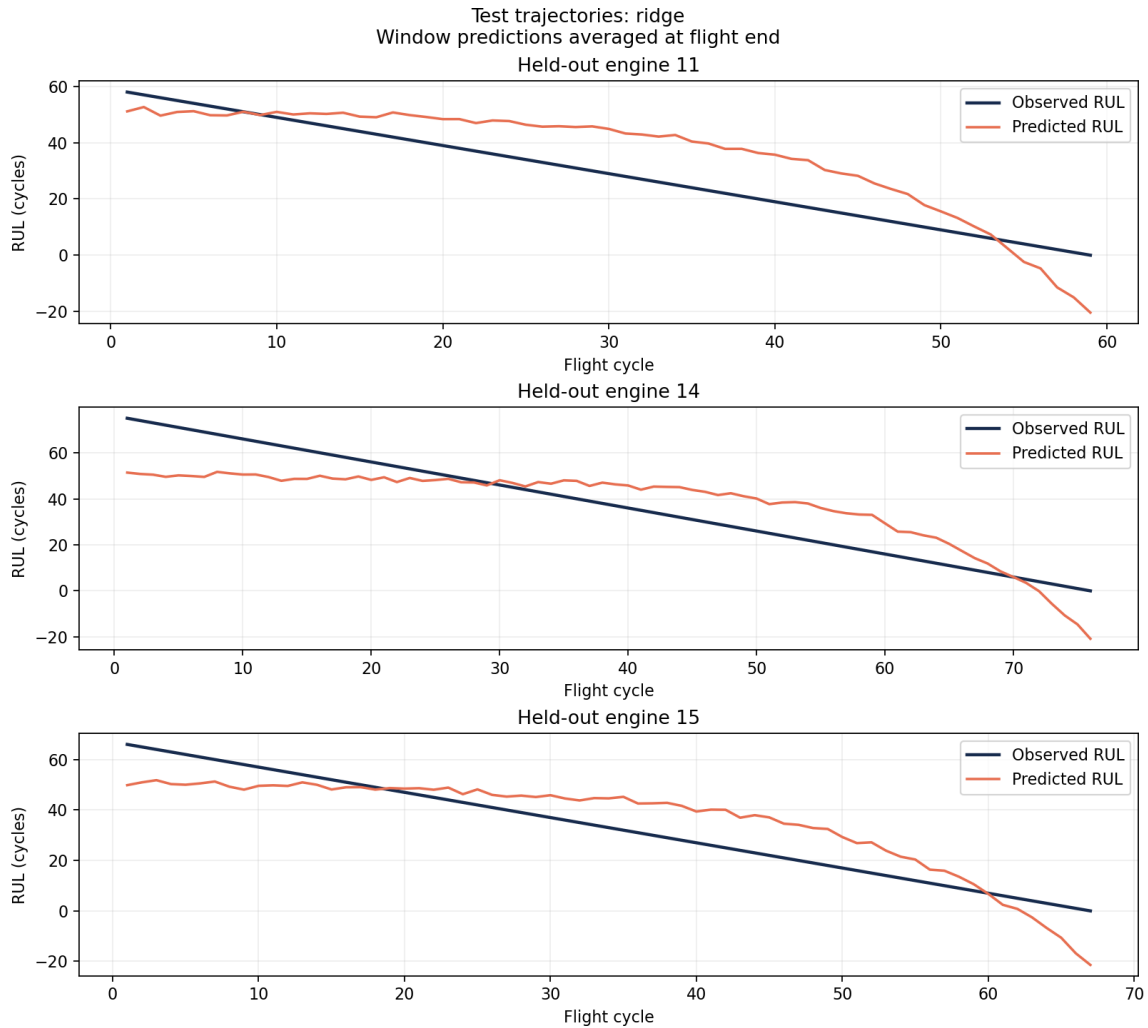


Figure 2. Test-engine RUL from Ridge, selected by development cross-validation. Each point averages window predictions from the same engine and flight cycle. Targets come from simulation, not operational aircraft observations.

Distribution shift and errors

All six development engines belong to class 3, so cross-validation did not directly validate transfer to classes 1 and 2. Ridge test RMSE was 11.417 cycles for engine 11, 12.296 for engine 14 and 10.116 for engine 15. Class-1 engine 14 had the largest error, but one test engine per class cannot establish a causal effect of flight class. Engine degradation histories and cycle counts also differ.

Interpreting the metrics

Flight-end averaging uses samples from the complete recorded flight and does not measure instantaneous online performance. Cycle and engine aggregation prevents longer flights with more windows from dominating the score. Raw-window and micro metrics are also retained in `metrics.json`; selection and the main conclusions here use macro cycle metrics.

Unconstrained Ridge produced negative RUL near end of life for all three test engines (15 of 202 cycles). Original results remain unclipped. A nonnegative policy was subsequently added and compared by development cross-validation. Because that decision followed test inspection, extension test results are not new independent generalization evidence. Negative remaining life is physically invalid and raw predictions are not suitable for operational decisions.

Parallel HDF5 Processing Experiment

Each run extracted the same 87,510 windows from 5,263,447 development rows. Six unique settings combined workers 1, 2, 4 and 8 with chunk sizes 1,024, 8,192 and 120,000 rows, with 3 measured repeats each. One complete pass warmed the OS cache; setting order was shuffled per repeat and each run used a fresh process.

Workers	Chunk rows	Median s	Min-max s	Speedup	Eff. %	RSS MiB	M rows/s
1	120,000	2.153	2.152-2.190	1.000	100.0	201.0	2.445
2	120,000	1.778	1.772-1.808	1.211	60.6	389.9	2.961
4	1,024	2.343	2.333-2.389	0.919	23.0	558.1	2.246
4	8,192	1.637	1.585-1.785	1.315	32.9	564.3	3.216
4	120,000	1.473	1.472-2.285	1.461	36.5	577.5	3.573
8	120,000	1.621	1.598-1.637	1.328	16.6	933.5	3.247

Speedup = median for one worker and 120,000-row chunks / setting median. Efficiency = speedup / worker count. RSS is the maximum sampled process-tree sum across repeats. Chunk experiments use the same baseline, so their speedup combines chunk and parallel effects.

Time median / min-max

Memory and speedup

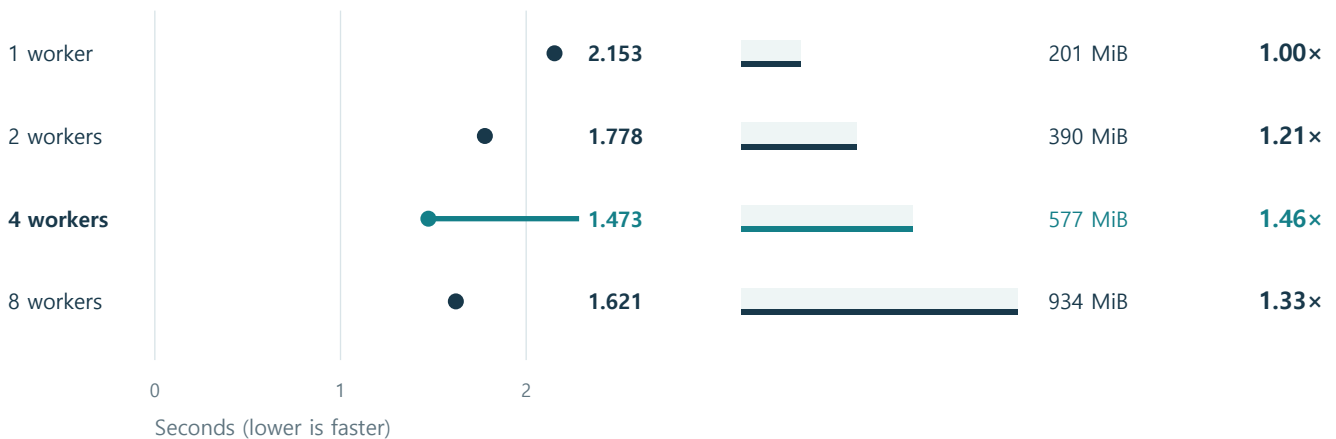


Figure 3. At 120,000-row chunks, dots show median time and lines span min-max; memory is peak RSS. Four workers were fastest, with one 2.285 s run. Memory increased with worker count.

Measurement scope and interpretation

The one-worker median was 2.153 s; eight workers took 1.621 s, giving speedup 1.328. More workers are not assumed to be faster. Spawning, dispatch, reads and result assembly all contribute, but this experiment does not isolate their individual bottleneck costs.

Timing includes HDF5 scanning, sensor reads, worker startup (including imports), feature calculation and assembly. It excludes initial imports in the parent process, result hashing and saving. Memory is process-tree RSS sampled every 100 ms and may double-count shared pages. The OS cache was not flushed; these are not cold-disk results. Ordered schemas, dtypes and row fingerprints matched in every run. All 114 code tests passed (output/test_results.txt).

Window and Prediction Policy Sensitivity

Six window/stride settings, Ridge and HGB, and raw/nonnegative policies yielded 24 development candidates. The nonnegative policy applies $\max(0, \text{window prediction})$ before flight averaging. Engine folds were anchored to the original 60/60 experiment. Every setting covers the same 446 development cycles. Only the development-selected candidate was refitted and then evaluated on test data.

Window / stride	Ridge raw	Ridge ≥ 0	HGB raw	HGB ≥ 0
30 / 15	11.371	10.779 *	11.891	11.898
30 / 30	11.375	10.795	11.866	11.872
60 / 30	11.379	10.799	11.883	11.888
60 / 60	11.383	10.822	11.775	11.781
120 / 60	11.393	10.830	11.735	11.742
120 / 120	11.402	10.866	11.754	11.760

Darker teal = lower error * selected by development CV Unit: cycles

Development macro engine cycle RMSE; * marks selection. Thirty-six CV model fits each support both prediction policies. Selection uses unrounded values; complete MAE, ranks and engine scores remain in CSV files.

Figure 4. Development RMSE heatmap for all 24 candidates; orange outlines the selected cell. Extension design followed original test inspection, so subsequent test scores remain exploratory.

10.779

Selected CV RMSE

Unit: cycles

30 / 15

Window / stride

Ridge · nonnegative

0.087

RMSE span, six settings

Nonnegative Ridge · cycles

Selected configuration

Selected: Ridge 30/15, nonnegative. Development RMSE/MAE: 10.779/8.795; reused-test RMSE/MAE: 10.591/9.021 cycles. Across six window settings, nonnegative Ridge CV RMSE spans only about 0.087 cycles. The minimum on this small grid is not a universal optimum.

1D CNN on Raw Sensor Windows

The CNN receives raw 18-channel $\times$ 60-sample windows instead of 90 summary features. Channel mean and standard deviation use engine/cycle weights from the current training partition. Three development folds and a full-development refit each run a fixed 12 epochs. Validation or test scores do not tune the training duration or hyperparameters.



Convolutions: kernel=5, padding=2 | Linear 32→1 | output $\times$ 100 | 4,081 parameters

Item	Actual setting
Training	Adam; epochs=12; batch=512; lr=0.001; target/100.0; seed=410
Device and size	NVIDIA GeForce RTX 4070 Ti SUPER; 4,081 parameters; PyTorch 2.5.1+cu121
Measured time	Total 74.71s; full-dev fit 21.03s; test prediction 0.21s

Softplus outputs are multiplied by 100 to recover nonnegative RUL. This range constraint does not establish accuracy or operational suitability. Optimization uses engine/cycle-weighted MSE on target/100. Exact numerical equality across hardware is not claimed.

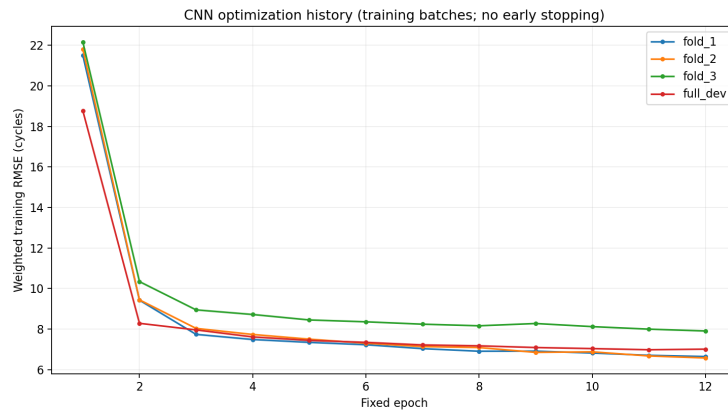
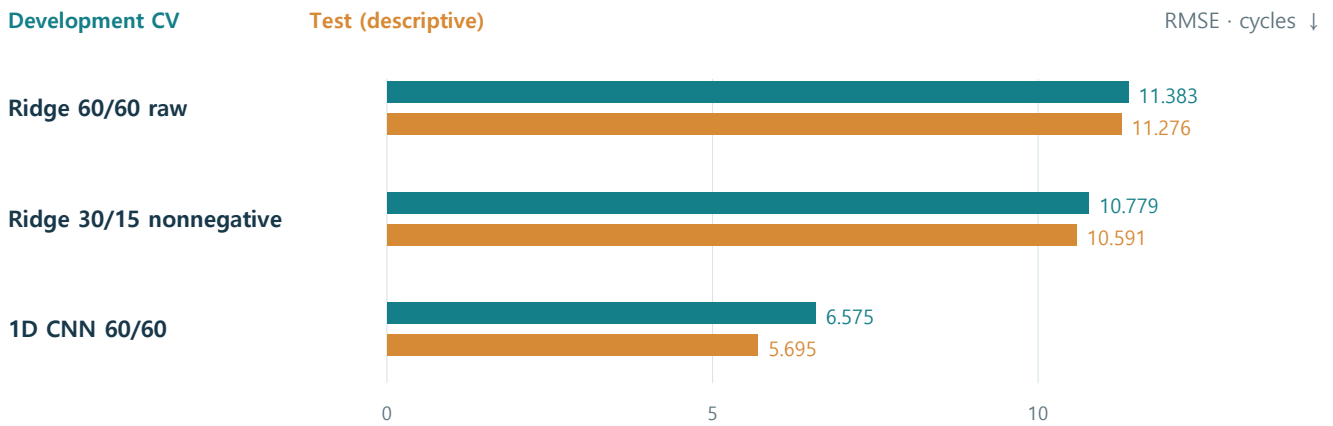


Figure 5. Weighted RMSE from training minibatches; not a validation curve or an early-stopping criterion.

Engine	Cycles	CNN MAE	CNN RMSE
11	59	6.186	7.478
14	76	4.671	6.151
15	67	2.705	3.458

CNN development macro RMSE was 6.575 and MAE 4.883. Reused-test macro RMSE was 5.695 and MAE 4.521 cycles. Its lowest development CV RMSE across candidates selected it as the final inference model.

Extension Comparison



These are repeated evaluations on the same test engines. Because extension scope was chosen after seeing original targets and results, differences are not improvements confirmed on a fresh independent test. Window/model/policy selection used development scores only.

Prediction diagnostics across 202 cycles

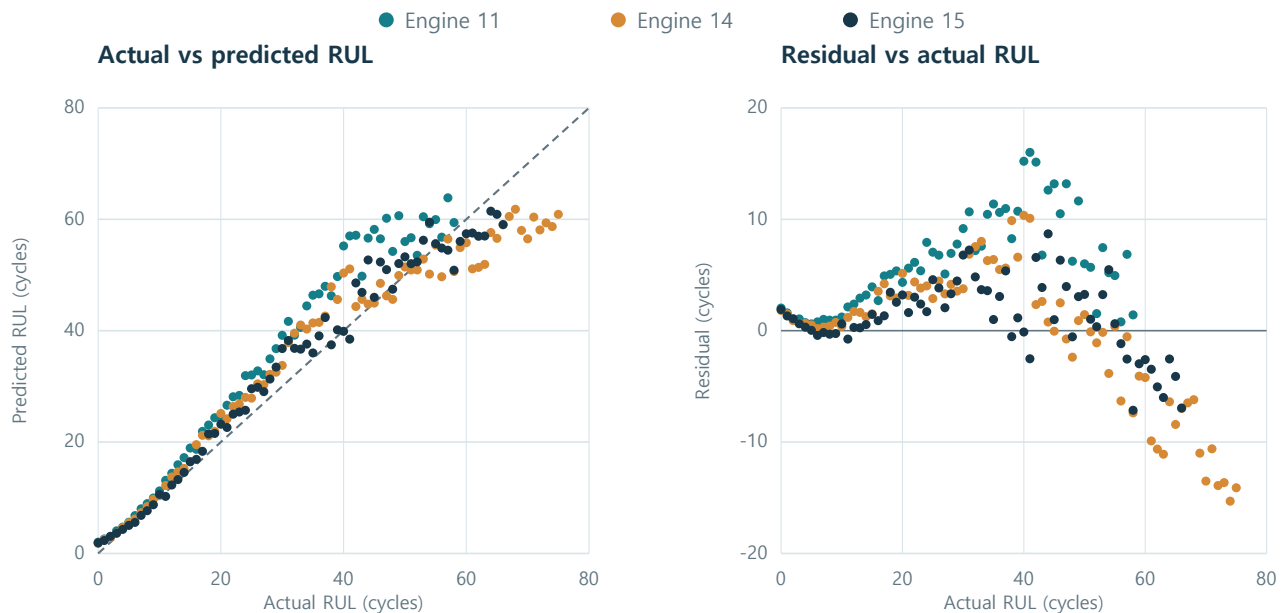


Figure 6. Top: development/test macro RMSE. Bottom: CNN actual-versus-predicted and residual (prediction minus target) plots. Each point is one flight cycle; color identifies the engine. The dashed diagonal denotes exact prediction; the horizontal line denotes zero error. Both vertical axes are in cycles. These are previously inspected test data, not fresh independent evidence.

Reused-test MAE: original Ridge 9.878, nonnegative Ridge 9.021, CNN 4.521 cycles. CNN was selected by the lowest development CV RMSE, not by test ranking.

Error Analysis and Execution Checks

True RUL was used for a post-hoc life-stage audit of the final CNN. These bins are not model inputs or selection criteria; they diagnose the same reused test data. Metrics average engine-level values within each bin, with all three test engines represented.

True RUL bin	Macro RMSE · cycles	MAE	Bias	Cycles
RUL ≤ 10	 0.956	0.818	+0.748	33
10 < RUL ≤ 30	 3.967	3.619	+3.594	60
RUL > 30	 7.186	6.216	+2.523	109

Bias is prediction minus target; positive values overestimate remaining life. Error was larger earlier in life (RUL > 30) than near end of life in this test. Small engine/bin samples do not establish guarantees for other operating histories.

Largest cycle errors

Engine	Cycle	Target	Prediction	Abs. error
11	18	41	57.008	16.008
14	2	74	58.693	15.307
11	19	40	55.212	15.212
11	17	42	57.135	15.135
14	1	75	60.896	14.104

Across 202 cycles there were 0 negative predictions; maximum absolute error was 16.008 cycles. Nonnegative outputs still have individual prediction errors. Both the original Ridge negatives and the remaining CNN errors are recorded.

Saved-model reuse checks

The final CNN is frozen in output/final/model.pt and model_card.json, retaining weights, channel order, training normalization and model hash. Target-free inference reproduced saved predictions for 20,796 test windows and 202 cycles without reading Y. A separate input with the target array removed also passed for 476 windows and 2 cycles. Inference does not require test labels.

Execution and artifacts

Use ./tools/complete_project.ps1 for the complete run and ./tools/rebuild_delivery.ps1 to rebuild documents/package from saved results. Predict a new target-free HDF5 file with ./run.ps1 -Stage predict -Dataset <path>. Retain the CNN checkpoint/card, window/cycle predictions, provenance/verification JSON, all CV candidates and original baseline results.

Completed Scope and Reproducibility

Limits of the experiment

DS02 has nine simulated engines and three test engines excluded from fitting. Millions of rows do not eliminate engine-level uncertainty. Reused-test extension comparisons remain exploratory. Development CV used for selection is not an independent nested-CV performance estimate. One engine per test class cannot identify class effects. No operational-aircraft or multinode-scaling claim is made.

Execution record and assignment deadlines

Implementation, experiments, inference and document generation were performed on 2 October. Remaining dates below are submission/presentation deadlines, not deferred implementation. All use Asia/Seoul.

Date	Item
10/02	Implementation, actual experiments, inference and bilingual documents
10/21	Submit one-page proposals in Korean and English
12/08	Five-minute presentation
12/16 23:59	Submit report and source code

Reproducibility record

Environment: Windows-11-10.0.26200-SP0; 8 physical cores, 16 logical CPUs, 31.1 GiB RAM. Configuration: configs/default.json. Provenance: data/raw/source_manifest.json. Feature audit: data/processed/feature_manifest.json. Results: output/results/metrics.json, benchmark_summary.csv and benchmark_runs.csv. Fitted models, prediction CSVs and figures are retained.

Set up the environment using README and requirements-tested.txt. Run `./run.ps1 -Stage download`, `./run.ps1 -Stage full`, then `./run.ps1 -Stage test`. For saved-model inference use `./run.ps1 -Stage predict -Dataset <HDF5 path>`; outputs go to output/predictions. Timings vary with hardware, cache and load.

References and data

- [1] NASA PCoE Data Set Repository. Entry 17: Turbofan Engine Degradation Simulation-2.
- [2] Arias Chao, M., Kulkarni, C., Goebel, K., & Fink, O. (2021). Aircraft Engine Run-to-Failure Dataset under Real Flight Conditions for Prognostics and Diagnostics. *Data*, 6(1), 5. doi:10.3390/data6010005.
- [3] Hao Li. N-CMAPSS DS02 third-party redistribution, Figshare version 1. doi:10.6084/m9.figshare.20436504.v1.